

Gulf Stream

The warm currents of the Gulf Stream make northern Europe's climate warmer than it would be otherwise.

How currents sink

When currents reach the cold polar oceans, some of the sea water freezes. When it does this, it leaves its salt behind. The salt mixes with the remaining water, making it saltier and heavier. This water then sinks toward the ocean floor and drives the currents that flow slowly through the ocean depths. Where these deep-water currents flow back up to the surface, scientists call it "upwelling."

Warm surface water flows in

Salt leaves the water when it freezes and makes the remaining water saltier and heavier

Cold, salty water sinks below the warm water and flows away slowly

Great Pacific Garbage Patch

Plastics and other garbage carried by currents collect within this slow-moving zone in the center of the North Pacific Gyre.

Deep water current

The deep current flowing across the basin of the Pacific begins to rise, warming up as it does so.

Southern oceans

Cold, dense water flows east across the deep ocean floor in the Antarctic, then heads north.

Friendly floaters

A cargo of plastic ducks lost in the Pacific in 1992 has helped scientists learn more about the speed and direction of ocean currents ever since. Some of the ducks drifted over 17,000 miles (27,500 km).

